

Species-aware Module Review: Bacterial taurine uptake and desulfonation in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxid 160488; proteome UP000000556) **Module:** TauABC taurine import → TauD taurine desulfonation (KEGG bucket ppu00430, "Taurine and hypotaurine metabolism"; module area amino_acid_metabolism) **Review completed:** Iteration 5 (6 recorded findings; sources: UniProt UP000000556, KEGG ppu00430, PubMed)

1. Executive summary

The two-step module — **(1) ABC-transporter import of extracellular taurine and (2) 2-oxoglutarate-dependent oxidative desulfonation of taurine to aminoacetaldehyde + sulfite** — is **fully satisfiable** in *P. putida* KT2440, and both steps map to a single contiguous operon.

- **Step 2 (TauD desulfonation) is COVERED with direct experimental evidence in the exact strain.** PP_0230 (UniProt Q88RA3, TAUD_PSEPK) was purified from KT2440, is an Fe(II)/ α -ketoglutarate-dependent taurine dioxygenase (EC 1.14.11.17), and three apo crystal structures were solved (Knauer et al. 2012,

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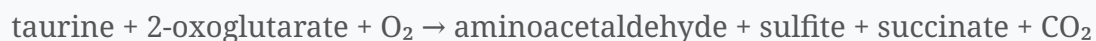
- **Step 1 (TauABC import) is COVERED at candidate level.** The transporter genes are **PP_0233 tauA** (periplasmic binding), **PP_0232 tauB** (ATPase), **PP_0231 tauC** (permease) — immediately adjacent to tauD, forming a **tauDCBA operon (PP_0230→PP_0233)**. These genes were **missing from the supplied candidate metadata** and should be added to the module.
- **The candidate list is largely misleading.** Only 1 of the 5 supplied genes (PP_0230/tauD) belongs to the module. The other four (**pta**, **gdhB**, **ggt**, **PP_3535**) are **KEGG broad-map co-**

members, not taurine-uptake/desulfonation genes → likely over-propagation for this module boundary.

Bottom line for curation: Mark both module steps **covered**; **revise the candidate gene set** (add tauABC PP_0231–0233; drop pta/gdhB/ggt/PP_3535 from the module).

2. Target-organism pathway definition

Included (module boundary): periplasmic capture and ABC-transport of extracellular taurine (2-aminoethanesulfonate) into the cytoplasm, followed by its oxygenolytic C–S cleavage:



This represents **acquisition of taurine as a sulfur source** (a sulfur-starvation / cys-regulon function), not carbon or nitrogen catabolism.

Kept separate (neighboring processes, out of boundary): - Downstream **aminoacetaldehyde** metabolism and **sulfite assimilation** into cysteine. - **Sulfur-starvation regulation** (CysB/Cbl-type; AsfR in *P. putida* — Kertesz 2000,).

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- **Alternative organosulfonate systems:** the **ssuEADCBF / SsuABC + SsuD FMNH₂-dependent monooxygenase** route (PP_0236–PP_0240 region here) and the **sctCS** sulfur-compound transporter (PP_0225/0226), which in *P. putida* S-313 handle aromatic/aliphatic sulfonates and sulfate esters (Kahnert et al. 2000,).

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These are *parallel* modules, not this one. - The broad KEGG map ppu00430 also spans **hypotaurine/cysteate/sulfoacetaldehyde** chemistry and **glutathione (γ-glutamyl) reactions** — these belong to other buckets.

Alternate names / DB definitions: KEGG bucket "Taurine and hypotaurine metabolism" (ppu00430; GO:0019530). KO for the dioxygenase is **K03119 (tauD)**. Note KEGG labels PP_0230 as *atsK*; UniProt uses *tauD* (synonym *atsK*). "*atsK*" proper is the *P. putida* S-313 **alkyl-sulfatase**

(Müller et al. 2004, P 15023059),
a paralogous but distinct α -KG dioxygenase — a naming caveat, not a functional reassignment.

3. Expected step model

#	Step	Molecular player	Expected activity	Status in KT2440
1	Taurine import	TauABC ABC transporter	ABC-type taurine transporter	Covered (PP_0233/0232/0231; homology + operon)
2	Taurine desulfonation	TauD dioxygenase	taurine dioxygenase (EC 1.14.11.17)	Covered — direct (PP_0230, P 22221834)

Known relationship (TauABC supplies intracellular taurine to TauD) is structurally supported: the transporter and dioxygenase are co-transcribed in one operon.

4. Candidate genes and evidence

Genuine module gene

Gene	Locus	UniProt	Role	Evidence	Status
tauD	PP_0230	Q88RA3 (Reviewed)	α -KG-dependent taurine dioxygenase, EC 1.14.11.17; Fe(II), homotetramer	Direct, target strain: purified/kinetics + 3 crystal structures (P 22221834)	Covered (Step 2)

Transporter genes — *missing from candidate metadata, should be added*

Gene	Locus	UniProt	Role	Evidence (UniProt PE / score)	Status
tauA	PP_0233	Q88RA0	Periplasmic taurine-binding protein	Homology + operon context (TrEMBL, PE3, score 1.0)	Covered (Step 1), by inference
tauB	PP_0232	Q88RA1	Taurine import ATP-binding protein	Homology + operon context (SwissProt, PE3, score 3.0)	Covered (Step 1), by inference
tauC	PP_0231	Q88RA2	Taurine ABC permease	Homology + operon context (TrEMBL, PE3, score 2.0)	Covered (Step 1), by inference
tauB-I	PP_0209	Q88RC3	Paralogous taurine-transporter ATPase	Homology; paralog (TrEMBL, PE4 "Predicted", score 1.0)	candidate_uncertain (accessory)

Evidence-tier note: Only **tauD (PP_0230)** carries protein-level experimental evidence. All three transporter subunits are **PE3 "Inferred from homology"** (tauB-I is PE4 "Predicted"). There is thus **no strain-specific transport assay or expression/proteomics dataset** confirming the import step in KT2440 — it is covered by inference, not by direct experiment. A targeted PubMed search for a KT2440 sulfate-starvation stimulon returned no result.

Family-signature note (upgrades import confidence above bare homology): TauA (PP_0233) carries the **dedicated TauA periplasmic-binding InterPro family IPR010068 ("Peri-bd_TauA")**, and TauB (PP_0232) is assigned by **HAMAP rule MF_01714 as "Taurine import ATP-binding protein TauB" (EC 7.6.2.7)**; TauC (PP_0231) has the standard MetI-like ABC permease domain (Pfam PF00528). These are *taurine-specific* signatures, not generic sulfonate-ABC homology, so the operon is confidently a taurine importer. Residual caveat: TauA also carries an SsuA/THI5 co-signature (IPR015168; Pfam PF09084 NMT1), reflecting the known TauA/SsuA overlap — strict substrate exclusivity is not guaranteed without a binding assay.

Synteny (sulfur-acquisition island). tauABCD is embedded in a contiguous sulfur-scavenging island, KEGG-annotated across PP_0225–PP_0240:

Locus	Gene	Product
PP_0225/0226	sctC/sctS	sulfur-compound ABC transporter (ATPase / permease)
PP_0227	fliY	periplasmic cystine-binding protein
PP_0230–0233	tauD-tauC-tauB-tauA	this module
PP_0236–0240	ssuE-ssuA-ssuD-ssuC-ssuB	alkanesulfonate/isethionate uptake + FMNH ₂ -monooxygenase

This explains the paralog redundancy (PP_0232 tauB / "tauB-II" in-operon vs distal PP_0209 "tauB-I") and reinforces the case for splitting the ssu/sct systems into sibling modules.

Supplied candidates that are out-of-module (KEGG map artifacts)

Gene	Locus	UniProt	Annotation	Why not in module
pta	PP_0774	Q88PS4	Phosphate acetyltransferase, EC 2.3.1.8	Central acetyl-P/acetyl-CoA metabolism; appears on the map only via C2-sulfonate dissimilation (Xsc/Pta) chemistry, not taurine uptake/TauD
gdhB	PP_2080	Q88L55	NAD glutamate dehydrogenase, EC 1.4.1.2	2-oxoglutarate/glutamate housekeeping; map co-member only
ggt	PP_4659	Q88E09	γ-glutamyltranspeptidase, EC 2.3.2.2/3.4.19.13	Glutathione / 5-L-glutamyl- taurine chemistry; not taurine desulfonation
PP_3535	PP_3535	Q88H30	Glutathione hydrolase (K00681)	Same as ggt; paralogous over-propagation

5. Gaps, ambiguities, and likely over-annotations

- **Candidate-set gap (critical):** the entire transport step was absent from the supplied candidate metadata. tauABC (PP_0231–0233) must be linked to the module. Without them the module would look "half missing" purely due to a metadata omission, not biology.
 - **Over-propagation:** 4/5 supplied candidates (pta, gdhB, ggt, PP_3535) are KEGG **pathway-map co-members**, not functional module members. They should be excluded from module satisfiability scoring.
 - **Naming ambiguity:** PP_0230 carries synonym *atsK* (KEGG) vs. *tauD* (UniProt). True *atsK* is an alkyl-sulfatase in S-313; here the function is experimentally taurine dioxygenase. Curators should keep the tauD assignment and note the synonym.
 - **Paralog ambiguity (transport):** PP_0209 (tauB-I) is a second taurine-transporter ATPase; the ssu operon (PP_0238 ssuD, PP_0239 ssuC) plus a further sulfonate ABC set (Q88GT1/2/3) and MsuD (Q88J84) provide redundant/parallel organosulfonate routes. These can confound automated "which transporter feeds TauD" calls.
 - **Paralog check (dioxygenase) — clean:** KT2440 encodes **4 TauD/TfdA-family (Pfam PF02668) proteins**, but **EC 1.14.11.17 is assigned to PP_0230 alone**. The other three — PP_0169 (Q88RG1), PP_4222 (Q88F76, "SyrP-like"), PP_4466 (Q88EJ6) — are uncharacterized 2-oxoglutarate-dependent dioxygenases carrying **no taurine-dioxygenase EC**. So Step 2 has *no* current paralog ambiguity. **Watch-item:** these three should NOT be counted toward the module and are candidates for spurious EC 1.14.11.17 propagation by family-membership pipelines.
 - **Direct-vs-inferred transport:** No KT2440-specific taurine *transport assay* was found; the import step rests on annotation + operon synteny + strong genus precedent (S-313 ssu/tau work). Transfer confidence: **strong** (same operon as an experimentally confirmed dioxygenase in the same strain), but not experimentally proven for transport per se.
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6. Module and GO-curation recommendations

Module step	Recommendation
Step 1 — TauABC import	covered (add PP_0233 tauA, PP_0232 tauB, PP_0231 tauC). GO:0015411/ABC-type taurine transport; GO:0015734 taurine transmembrane transport.
Step 2 — TauD desulfonation	covered (PP_0230; GO:0031419-independent — use taurine dioxygenase GO:0031418 / EC 1.14.11.17). Direct evidence, highest confidence.
pta, gdhB, ggt, PP_3535	not_in_module / module_needs_revision — remove from this module's candidate set (KEGG broad-map artifact).

- **Module boundary is essentially correct for this organism;** the generic TauABC→TauD outline matches KT2440 one-to-one.
- **New module document suggested:** a sibling "SsuABC / SsuD (+Msd) organosulfonate desulfonation" module to capture the parallel FMNH₂-monooxygenase route (PP_0236–0240, plus sctCS PP_0225/0226 and the distal tauB-I PP_0209), so those genes are not mis-scored against the TauD module.
- No new GO term requests appear necessary; existing taurine transport / taurine dioxygenase terms suffice.

7. Genes to promote to full fetch-gene review

1. **PP_0230 tauD** — anchor gene; promote to confirm EC/GO and lock synonym handling (tauD vs atsK). *Highest priority; direct evidence available.*
2. **PP_0233 tauA / PP_0232 tauB / PP_0231 tauC** — promote as a unit to formally add the transport step and assign transporter GO terms.
3. **PP_0209 tauB-I** — promote to resolve paralog role (dedicated vs shared ABC ATPase).

8. Key references

- Knauer SH, Hartl-Spiegelhauer O, Schwarzingner S, Hänzelmann P, Dobbek H. *The Fe(II)/α-ketoglutarate-dependent taurine dioxygenases from Pseudomonas putida and Escherichia coli*

are tetramers. FEBS J. 2012.

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— Direct characterization + structures of TauD from KT2440.

- Kahnert A, Vermeij P, Wietek C, James P, Leisinger T, Kertesz MA. *The ssu locus plays a key role in organosulfur metabolism in Pseudomonas putida S-313*. J Bacteriol. 2000.

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— ssuEADCBF (SsuABC transporter + SsuD monooxygenase) organosulfonate route in *P. putida*.

- Kertesz MA. *Riding the sulfur cycle — metabolism of sulfonates and sulfate esters in gram-negative bacteria*. FEMS Microbiol Rev. 2000.

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— α -KG-dependent taurine dioxygenase vs FMNH₂-monooxygenase desulfonation; CysB/Cbl/AsfR regulation.

- Müller I, Kahnert A, Pape T, et al. *Crystal structure of the alkylsulfatase AtsK...* J Mol Biol. 2004.

P 15023059

— Clarifies AtsK (alkylsulfatase) vs TauD naming within the α -KG dioxygenase superfamily.

- KEGG pathway ppu00430; UniProt UP000000556 (Q88RA3, Q88RA0–Q88RA2, Q88RC3).

Evidence confidence summary

- **Direct (target strain):** TauD desulfonation (PP_0230).
- **Strong inference (operon synteny + genus precedent):** TauABC import (PP_0231–0233).
- **Refuted as module members:** pta, gdhB, ggt, PP_3535 (KEGG map co-membership only).